

A Literature Review on AI-Driven Security and Performance Assessment for Websites and APIs

Abstract

Modern web applications rely heavily on websites and RESTful APIs, making security and performance critical aspects of software quality. Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs) have introduced new approaches for vulnerability prioritization, automated website assessment, API security analysis, and intelligent decision support. This review analyses six recent studies to examine current developments in these research areas, with particular emphasis on AI-based vulnerability prioritization, website and API security assessment, performance evaluation, and AI-assisted website quality analysis. The review highlights the strengths and limitations of existing approaches, identifies key research gaps, and discusses the need for an integrated framework that combines security assessment, performance optimization, and explainable AI. Based on these findings, the study motivates the development of an AI-driven decision support system capable of providing comprehensive website and API security assessment within a unified platform.

Keywords: *Artificial Intelligence, Website Security, REST API Security, Vulnerability Prioritization, Performance Optimization, Decision Support System*

1. Introduction

With the rapid advancement of digital technologies, websites and RESTful APIs have become the backbone of modern applications across industries such as banking, healthcare, education, e-commerce, and government services. As these applications continue to evolve, they process large volumes of sensitive information and support millions of users every day. This increasing dependence on web-based systems has made security, performance, and reliability critical factors in delivering a safe and efficient user experience. At the same time, the growing complexity of web applications has introduced new challenges, including security vulnerabilities, insecure APIs, performance bottlenecks, and the difficulty of analysing large volumes of security and performance data.

To address these challenges, researchers have proposed a wide range of techniques for website assessment, vulnerability management, API security analysis, and performance evaluation. Traditional approaches mainly focus on identifying vulnerabilities or measuring website performance independently, while recent studies have introduced Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs) to automate vulnerability prioritization, security analysis, and intelligent report generation. These advancements have significantly improved the efficiency of website assessment and decision-making. However, most existing solutions are designed to solve specific problems rather than providing a unified view of a website's overall security and performance.

For this review, six recent research papers were analysed to understand the latest developments in AI-based vulnerability prioritization, website and API security assessment, performance evaluation, and AI-assisted website quality analysis. Among these, three studies focusing on AI-driven vulnerability prioritization, automated website and API assessment, and LLM-assisted website quality analysis serve as the primary foundation for the proposed framework, while the remaining studies provide supporting insights into current research trends and existing challenges.

The objective of this review is to examine the strengths and limitations of existing approaches, identify the major research gaps, and understand how recent developments have shaped intelligent website assessment systems. To provide a structured discussion, the reviewed studies are categorized into three major research domains: **AI-based vulnerability prioritization, website security, API assessment and performance evaluation, and AI-assisted website quality analysis**. The findings from these domains form the basis for the comparative analysis, research gap identification, and the evolution of the proposed **AI-Driven Decision Support System for Integrated Website and API Security and Performance Optimization**.

2. Review of Existing Literature

2.1 AI-Based Vulnerability Prioritization

The increasing number of software vulnerabilities reported each year has made vulnerability prioritization an important aspect of cybersecurity. While vulnerability scanners can detect a large number of security issues, determining which vulnerabilities should be addressed first remains a major challenge. Traditional prioritization techniques primarily rely on the Common Vulnerability Scoring System (CVSS), which assigns severity scores based on predefined technical parameters. Although CVSS provides a standardized approach for evaluating vulnerabilities, it often fails to consider factors such as exploitability, business impact, and the operational context of an application, leading to inefficient prioritization of security risks.

Recent research has attempted to overcome these limitations by integrating Artificial Intelligence (AI) and Machine Learning (ML) into vulnerability management. AI-based approaches analyse contextual information along with vulnerability characteristics to improve risk assessment and prioritize security issues more effectively. Machine Learning models have also been used to predict vulnerability severity, automate risk ranking, and support remediation planning. These methods reduce manual effort and enable security teams to focus on vulnerabilities that pose the greatest threat to an organization.

Despite these advancements, existing vulnerability prioritization techniques remain focused primarily on security scoring and remediation planning. They provide limited support for correlating vulnerability data with website performance, API security, or overall application health. As a result, developers often need to rely on additional tools to obtain a complete understanding of system security. This highlights the need for a more integrated approach that combines vulnerability prioritization with broader website assessment and intelligent decision support.

2.2 Website Security, API Assessment and Performance Evaluation

Modern web applications depend heavily on websites and RESTful APIs to deliver services and exchange information. As these applications continue to grow in complexity, maintaining both security and performance has become equally important. A secure website with poor performance can negatively affect user experience, while a high-performing application with security vulnerabilities remains vulnerable to cyber threats. Therefore, recent research has focused on developing automated techniques that can evaluate website security, API behaviour, and overall system performance.

Automated website assessment has become an effective approach for identifying security weaknesses and evaluating application quality without extensive manual intervention. Network-layer testing techniques analyse HTTP requests and responses to examine how websites and APIs behave during execution. These techniques help identify issues related to authentication, communication protocols, response handling, and configuration errors, allowing developers to detect vulnerabilities under real operating conditions. Compared to traditional testing approaches, automated assessment provides faster analysis and improves the reliability of security evaluation.

Along with website assessment, API security has gained significant attention because APIs are now responsible for handling most communication between applications and services. Common security issues such as broken authentication, improper authorization, insecure data exposure, and injection attacks continue to pose serious risks to modern web applications. Existing studies have explored static analysis, dynamic testing, and automated vulnerability detection techniques to improve API security. While these methods effectively identify API-specific vulnerabilities, they are generally implemented as standalone solutions and do not provide a complete assessment of the associated website.

Performance evaluation is another important factor in determining the overall quality of a web application. Current approaches measure parameters such as page load time, response latency, throughput, and resource utilization to understand system efficiency under different workloads. Although these techniques provide valuable insights into application performance, they are often evaluated independently from security assessment. As a result, developers receive separate reports for security, API behaviour, and performance, making it difficult to understand the overall health of the application or prioritize improvements effectively.

Overall, the existing literature demonstrates significant progress in automated website assessment, API security analysis, and performance evaluation. However, these approaches remain fragmented, requiring multiple tools and separate analyses to obtain a complete picture of website quality. This highlights the need for an integrated framework capable of assessing security, API behaviour, and performance together while providing meaningful insights for developers and security analysts.

2.3 AI-Assisted Website Quality Analysis

The rapid advancement of Artificial Intelligence (AI) has significantly influenced the way website quality and security are evaluated. In recent years, Large Language Models (LLMs) have emerged as promising tools for assisting developers in understanding security reports, identifying potential issues, and generating human-readable recommendations. Unlike traditional assessment tools that primarily produce technical outputs, LLM-based systems can interpret complex findings and present them in a more understandable and actionable form. This makes security analysis more accessible, particularly for developers who may not have extensive cybersecurity expertise.

Recent studies have explored the use of AI-powered plugins and LLMs to automate website quality analysis and simplify decision-making. These systems are capable of summarizing website assessment results, explaining detected vulnerabilities, and suggesting possible remediation strategies. Such intelligent assistance reduces the time required to analyse lengthy security reports and enables developers to focus on resolving critical issues more efficiently. The integration of AI also improves the overall usability of security assessment tools by transforming technical information into meaningful insights.

Despite these advantages, current AI-based approaches still face several challenges. Large Language Models may occasionally generate inaccurate or inconsistent recommendations, commonly referred to as hallucinations, which can affect the reliability of automated analysis. Moreover, existing solutions mainly focus on interpreting assessment reports rather than integrating vulnerability prioritization, website security, API analysis, and performance evaluation into a unified framework. These limitations indicate that while AI has strong potential to support cybersecurity, its capabilities can be further enhanced by combining intelligent decision support with comprehensive website assessment techniques.

3. Comparative Analysis

The reviewed studies collectively demonstrate the growing use of Artificial Intelligence in improving website security, API assessment, vulnerability prioritization, and performance evaluation. Earlier research primarily focused on identifying vulnerabilities and assigning severity scores using traditional methods such as CVSS, while more recent studies have incorporated Machine Learning to improve risk prioritization and support remediation planning. Similarly, website assessment has evolved from conventional security testing to automated evaluation of website behaviour, API communication, and application performance. The introduction of Large Language Models has further enhanced this process by providing intelligent explanations and recommendations, making security reports easier to understand and interpret.

Although these studies have significantly advanced their respective research areas, each of them addresses only a specific aspect of website assessment. AI-based approaches mainly concentrate on vulnerability prioritization, while website and API assessment techniques focus on identifying security weaknesses without considering overall system performance. Likewise, performance evaluation methods measure application efficiency but are generally independent of security analysis. AI-assisted website quality analysis improves the interpretation of assessment results but does not combine information from different domains to support comprehensive decision-making.

Overall, the literature indicates that website security, API security, performance evaluation, and AI-driven analysis are still treated as separate research areas. As a result, developers often depend on multiple tools and manually combine the findings to understand the overall health of a web application. This fragmented approach increases the complexity of security management and highlights the need for a unified framework that integrates these complementary techniques into a single intelligent decision support system.

4. Research Gap Analysis

The review of existing literature highlights that although significant progress has been made in website security assessment, API security, vulnerability prioritization, performance evaluation, and AI-assisted analysis, several important challenges remain unresolved. Most of the existing approaches are designed to address specific problems independently, resulting in fragmented analysis and limited support for comprehensive website evaluation. Developers often rely on multiple tools to assess different aspects of a web application, making the overall assessment process time-consuming and difficult to manage.

One of the major research gaps identified is the absence of an integrated framework that simultaneously evaluates website security, API security, and performance. Current solutions generally focus on either security assessment or performance evaluation, without considering the strong relationship between these two aspects. Similarly, vulnerability prioritization techniques continue to depend largely on static scoring mechanisms, with limited consideration of contextual information, exploitability, and real-world system impact. This can lead to inaccurate prioritization and inefficient allocation of security resources.

Another important limitation is the restricted use of Artificial Intelligence for holistic decision-making. Although AI and Machine Learning have been successfully applied to vulnerability prioritization and website quality analysis, their integration across multiple assessment domains remains limited. In addition, many AI-based solutions operate as black-box models, providing recommendations without sufficient explanation. This lack of transparency reduces user confidence and makes it difficult for developers to understand the reasoning behind automated decisions.

The literature also reveals limited support for continuous monitoring, adaptive performance evaluation, and large-scale real-world validation. Furthermore, none of the reviewed studies proposes a unified risk scoring mechanism that combines both security and performance findings into a single assessment. These limitations clearly indicate the need for an intelligent framework capable of integrating multiple assessment techniques while providing explainable, context-aware, and actionable recommendations for developers.

5. Evolution Towards the Proposed Framework

The review of existing literature demonstrates that significant progress has been made in individual areas such as vulnerability prioritization, website security assessment, API security analysis, performance evaluation,

and AI-assisted website quality analysis. Each of these research areas contributes valuable techniques for improving web application security and reliability. AI and Machine Learning have enhanced vulnerability prioritization, automated testing has strengthened website and API assessment, and Large Language Models have simplified the interpretation of security reports by providing intelligent recommendations. These advancements collectively represent an important step toward more efficient and automated website assessment.

However, the comparative analysis and identified research gaps indicate that these solutions continue to operate independently. Existing approaches generally focus on solving a single problem and require developers to use multiple tools to obtain a complete understanding of a website's security and performance. The absence of an integrated framework capable of combining website security assessment, API analysis, performance evaluation, intelligent vulnerability prioritization, and explainable decision support limits the effectiveness of current cybersecurity practices.

Motivated by these observations, this review proposes the need for an integrated framework that brings together the strengths of existing research while addressing their limitations. The proposed framework, **AI-Driven Decision Support System for Integrated Website and API Security & Performance Optimization**, aims to provide a unified platform for evaluating website security, analysing API vulnerabilities, assessing performance metrics, prioritizing security risks using AI, and generating explainable recommendations to support developer decision-making. By integrating these capabilities into a single system, the proposed framework seeks to simplify website assessment, improve vulnerability management, and enhance the overall security and performance of modern web applications.

The proposed framework represents a natural progression of the existing research rather than a replacement for current approaches. It builds upon the contributions of previous studies by integrating their strengths into a comprehensive decision support system capable of providing a more holistic, intelligent, and practical solution for website and API security assessment.

By integrating these complementary research areas into a single intelligent framework, the proposed system aims to bridge the gap between isolated security assessment tools and practical AI-driven decision support for modern web applications.

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